

Mercy Naserian Mang'oka

Personal Information

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Location: Nairobi, Kenya

Professional Summary

Motivated Software AI Engineer and full-stack developer with hands-on experience building production-grade AI-powered applications, RESTful APIs, and scalable data pipelines. Proficient in Python, LLM integration, RAG architectures, and multi-agent AI systems. Experienced in deploying cloud-based solutions on AWS and developing user-focused platforms that combine AI reasoning with robust backend engineering. Passionate about the responsible development of frontier AI and committed to contributing to high-impact, mission-critical projects.

Skills

- **AI Engineering:** LLMs, RAG (Retrieval-Augmented Generation), Multi-Agent Systems, Prompt Engineering, LangChain
- **Python Development:** REST API Design, Flask, Secure Coding, ETL Pipelines
- **Cloud & Infrastructure:** AWS, API Development, CI/CD Practices
- **Full Stack Development (MERN):** MongoDB, Express.js, React.js, Node.js
- **Frontend:** HTML, CSS, JavaScript, TypeScript, Next.js
- **Databases:** MongoDB, PostgreSQL, MySQL, SQL
- **AI & Data:** Machine Learning, TF-IDF, User Behavior Analysis, AI Engineering
- **Tools:** Git, GitHub, Version Control

- **Soft Skills:** Communication & Teamwork, Technical & Non-Technical Stakeholder Engagement

Experience

Software Development Intern — Kenya Revenue Authority (KRA)

January 2026 – March 2026

- Collaborated with the IT department to design and develop software solutions for government-facing digital systems.
- Applied full-stack and API development principles to enhance internal platforms, adhering to secure coding standards.
- Participated in SDLC processes including requirements gathering, development, testing, and deployment.
- Communicated progress and technical concepts clearly across both technical and non-technical teams.

Projects

AI-Powered E-commerce Recommendation System

- **LLM & AI Integration:** Built a full-stack e-commerce platform with an AI recommendation engine leveraging TF-IDF and user behavior analysis — demonstrating core RAG and retrieval pipeline concepts.
- **Backend & API:** Designed and deployed RESTful APIs in Python to serve personalized product recommendations at scale, with authentication, cart management, and session handling.
- **Agentic Workflow:** Implemented an intelligent recommendation interface that adapts dynamically to user interactions, reflecting multi-agent and context-aware AI design principles.

Automotive Valuation & Tax Compliance Platform

- **API Architecture:** Engineered a robust RESTful API using Python (Flask) to handle complex taxation logic and CRSP data management for a government-aligned compliance platform.
- **Data Pipeline:** Built optimized ETL-style data processing pipelines to serve thousands of vehicle records with server-side pagination and query filtering.
- **Frontend Engineering:** Developed a responsive dashboard using React and Tailwind CSS with a real-time import tax calculator.

Smart Match Job Portal

- Designed and built a job-matching system using automated skill-based matching algorithms — a foundation applicable to agentic AI workflows and intelligent recommendation systems.
- Implemented user registration, job postings, and a dynamic recommendation interface using full-stack MERN architecture.

Education

Kabarak University — Bachelor of Science in Information Technology

Coursework Completed: 2025 | Expected Graduation: 2026

Relevant coursework: Programming, Networking, Databases, Software Development, AI Fundamentals.

Professional Training & Certifications

Power Learn Project (PLP) Academy — Full Stack Development (MERN)

Completed: 2025

Hands-on training in React, Node.js, Express, MongoDB, Python, Database Management, Prompt Engineering, and API development — directly aligned with LLM and AI engineering practices.

KCA University — Web Development Short Course

Completed: 2024

Practical skills in HTML, CSS, and JavaScript.

Languages

- English — Fluent
- Kiswahili — Fluent